A Low-cost, Scalable Mechatronics Course Centered on Tinker-and-Build Labs and Project-based Learning

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Course material: https://tiny.cc/mechatronix

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Two Challenges in Mechatronics Education

- High Cost of Setting up Labs
- Lack of Integration of Lectures and Laboratories

Our attempt to lower the barrier for Mechatronics Education

All material is at: https://tiny.cc/mechatronix

Overview

- Offering:
  - Core (senior): 90-120 students (3 times)
  - Elective (senior): 20-30 students (2 times)
- Interaction: Two-50 minutes; 2-hour laboratory per week
- Assessment: HW, Labs, 2 Exams, Project.

Lectures/Labs

- Lectures: Capacitors/Resistors/Transistors/Op-amps/Logic Gates/Flip-Flops/Sensors/Actuators
- Labs: 11 labs + 1 makeup
  - Arduino IDE (C-based)
  - TinkerCAD (Circuit simulator/CAD)

Schedule: 16 week semester

#	Lectures	Labs
1	Basic Electrical Components; Kirchhoff's Current/Voltage Law	No Lab **11 Labs**
2	Analyzing Circuits; Practical Electronics; Characteristics of a Good Measurement System **Lectures + HW**	Lab 1 – Tinker CAD
3	Diodes (Junction, Zener, LED, Photodiode)	Lab 2 – Resistors (Voltage/Current Divider)
4	Bipolar Junction Transistor (Theory & Examples)	Lab 3 – Capacitors (High/Low Pass Filter)
5	Amplifiers (Op-amp Basics, Inverting/Non-inverting, Applications)	Lab 4 – Diodes
6	Op-amp Applications (Instrumentation, Comparator, Sample & Hold); Real Op-amp Specs	Lab 5 – Bipolar Junction Transistor
7	Exam 1 **Exam 1**	Lab 6 – Integrated Circuits
8	Digital Circuits; Logic Gates; Boolean Algebra	Lab 7 – Soldering
9	Logic Gates (Problems & Boolean Expressions)	Lab 8 – Arduino (Hardware)
10	Sequential Logic; Flip Flops; Applications	Lab 9 – Design and 3D Printing
11	Counters, Decoders, Timers, Data Sampling; Project: Interactive Animatronics	Lab 10 – Sensors and Actuators 1
12	Data Quantization; A/D and D/A Converters	Lab 11 – Sensors and Actuators 2
13	Sensors (Emitter/Detector, Ultrasonic, RFID, Encoders); Measurement (Strain, Force, Temp, Vibration)	Makeup Lab / Project week 1 **Course Project**
14	Actuators (Solenoids, DC Motors, Stepper, Brushless); Hydraulics & Pneumatics	Project week 2
15	Exam 2 **Exam 2**	Project week 3
16	Public Demonstration of project **Project Demo**	

Scaffolding: R-C circuits

2.1 Function Generator and Oscilloscope

2.1.1 (20 points) Simulation in TinkerCAD

1) Learn new equipment in simulation

Watch this video on using the TinkerCAD Function generator and Oscilloscope. https://youtu.be/wfvYz1L0Qkk. Connect the Function generator to the oscilloscope. Play with the frequency, amplitude, DC offset, and function on the function generator and note the output on the oscilloscope.

Use of videos by other YouTubers

1. Generate a square wave of frequency of 10 Hz and amplitude of 10 V. Show the output on the oscilloscope.

2. Now add an offset of 5V. What happens to the output?

Show your work to the teaching assistant.

2) Redo in hardware

2.1.2 (20 points) Hardware

Repeat the above TinkerCAD exercise but using the oscilloscope/function generator in the lab. You could also use 2 Hantek; one as a function generator and one as an oscilloscope. Here is a short video showing how to connect and use a function generator/oscilloscope https://youtu.be/jWDJeiH6veQ. Show your work to the teaching assistant.

3) Build/Measure RC circuit in hardware

2.2 RC Filters

RC circuits are composed of resistors and capacitors. RC circuits can be used to filter a signal by blocking certain frequencies and passing others. We will study the simplest RC circuit consisting of one resistor and one capacitor. RC circuit can be used as a low pass filter (remove high frequency signals) as well as high pass filter (remove low frequency signals).

Scaffolding: Sensors/Actuators

1) Learn about a sensor

2.4 (20 pts) Ultrasonic sensor, a distance sensor

This webpage gives theory behind ultrasonic sensor and how to hook it up: https://howtomechatronics.com/tutorials/arduino/ultrasonic-sensor-hc-sr04/. Hook up the sensor to the arduino and calibrate it to read the distance. Demonstrate that the sensor is able to measure the distance and show it to the TA.

Use of material created by others

2.5 (20 pts) DC motor

This webpage shows how to interface and program the DC motor. https://create.arduino.cc/projecthub/ryanchan/how-to-use-the-l298n-motor-driver-b124c5. Demonstrate that you are able to spin a single DC motor back and forth using the H-bridge and various speeds.

2) Learn about an actuator

2.6 (30 pts) Application as a backup sensor

Create a prototype of an automated backup system for an electric car. The DC motor is the proxy for the rear wheel. The DC Motor should be freely spinning. Then, as the distance between the sensor and the obstacle decreases, the speed of the DC motor should decrease till it reaches zero when speed reaches a critical value. Demonstrate the prototype to the teaching assistant.

3) Solve a real-world application using learned knowledge

Costs

One time: $2500 for 10 groups; soldering station, oscilloscope, function generator, small consumables, resistors, diodes

Personnel Cost: 1 TA for 10 groups

Per semester: $65/student

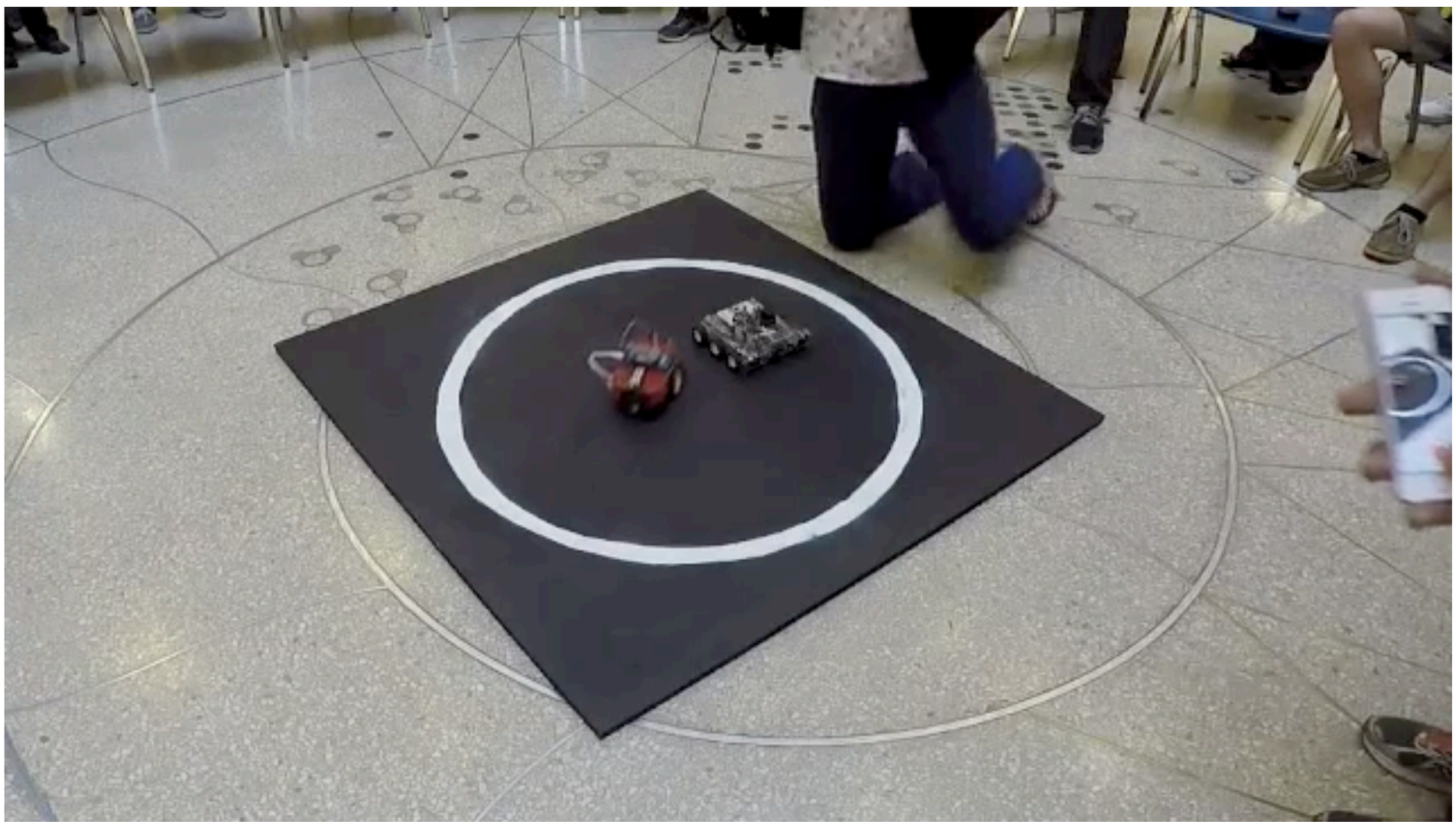
Item	Which lab	Qty	Price	Total
ON/OFF switch	Arduino	30	0.95	28.5
TIP31C power dio	BJT	30	0.73	21.9
Photo interuppte	BJT	30	0.35	10.5
10 uF capacitor	Capacitors	30	0.5	15
LM741 op-amp	IC	30	0.9	27
555 timer	IC	30	1	30
Zener diode	Diode	30	0.35	10.5
L298 H-bridge	Sensors/Actuators	30	2.2	66
Barrel jack	soldering	30	0.77	23.1
DC motor	project	30	2.95	88.5
Power adaptor	project	30	8.95	268.5
Perf board	soldering	2	8	16
Elegoo UNO R3 Super Starter Kit		30	45	1350
			Total	1955.5
	Students	30	Per Student	65

Project

- Week 13: Literature Review, Brainstorm Ideas, Present to Instructor
- Week 14: Build the support structure/3D print
- Week 15: Interface Sensors / Motors
- Week 16: Public demonstration
- Report: https://www.instructables.com/
- Grading: Week13-15 (40 pts), Week 16 (30 points), Report (30 points)

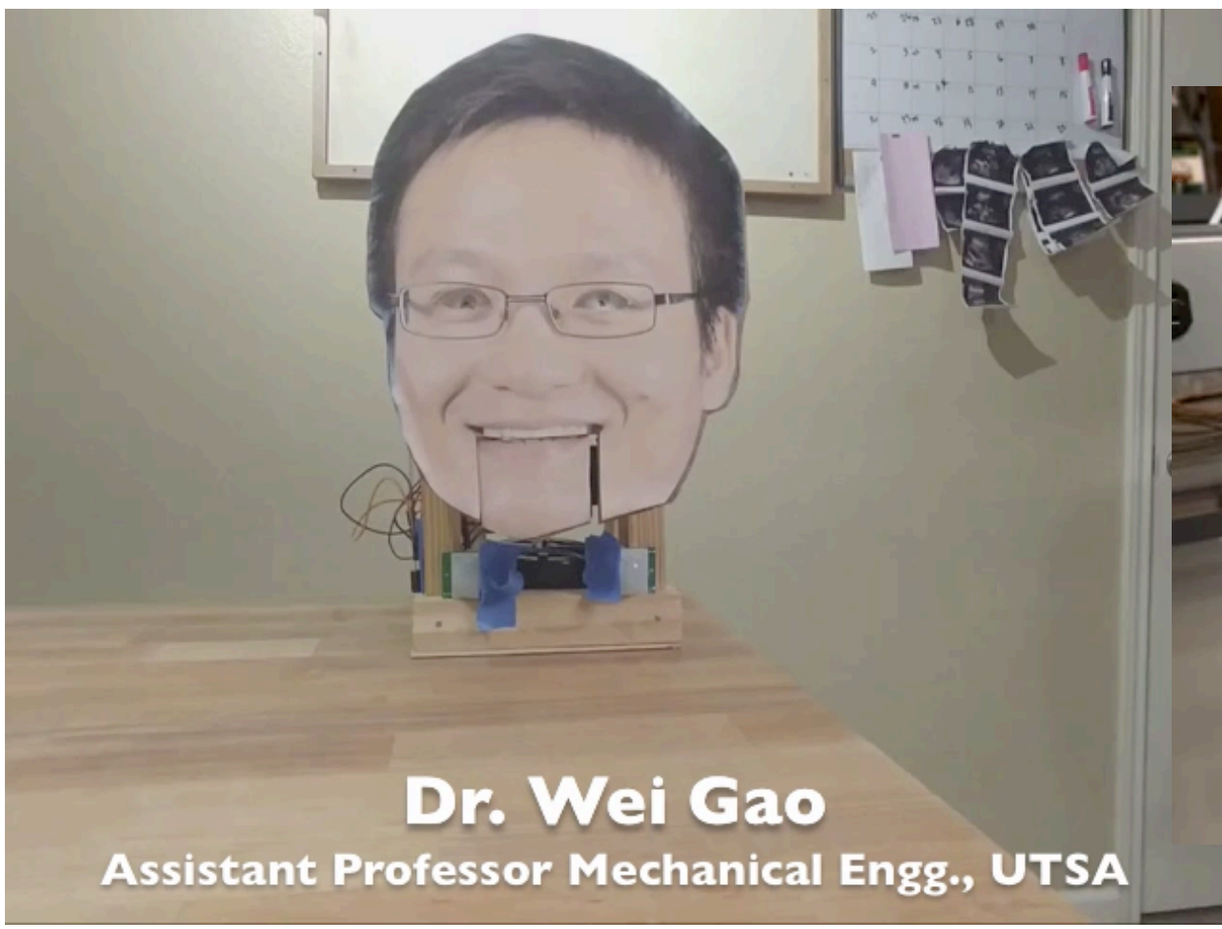
Project: Sumo-Bot

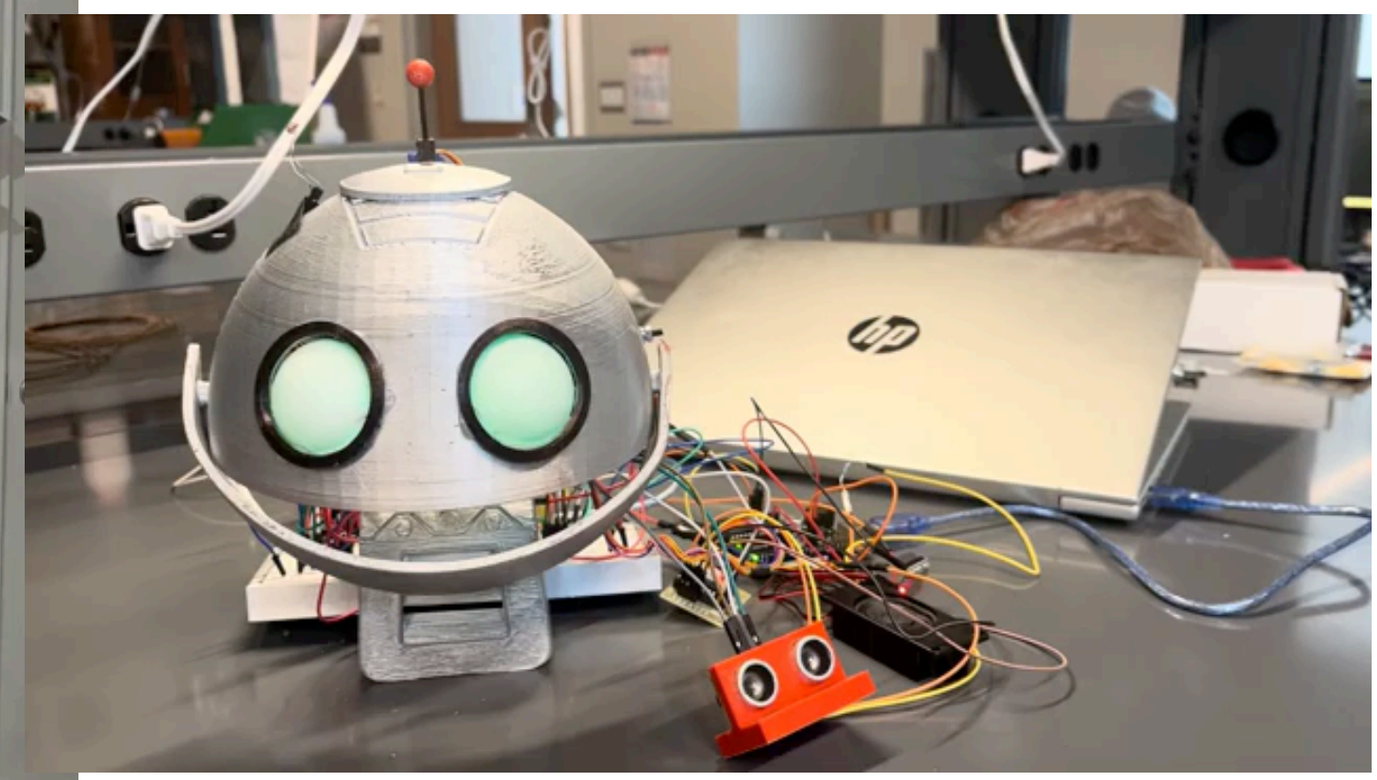
- A robot that can push the opponent out
- Size/Weight/Cost/Power constraint



Project: Animatronics

- Robot Face that interacts with users
- Uses one motor/two sensors/3D-printed part





Discussion

- Course Placement: Junior at 300-level
- Integrate real-world examples: 1) Videos; 2) Look up datasheet/components
- ABET: (1*) engineering problem solving; (2*) engineering design under constraints; (3*) effective technical communication; (4) ethical and societal considerations; (5*) teamwork and leadership; (6*) experimentation and data analysis; and (7*) lifelong learning.
- TinkerCAD: Circuit Simulator / CAD Design (It's free)
- Personnel: 1 TA for 10 groups (1 group = 1 - 4 people)
- Learn by Trial-and-Error: Reflects real-world experience

Conclusion

- Mechatronics course material
  - Low Cost: $65 per student
  - Scalable: Cost/Space scales linearly with students
  - Adaptable:
    - 1) Modify for quarter/semester system
    - 2) Modify for major (e.g., Civil, Biomedical)
    - 3) Upgrade for Robotics

Thank you

- All software is free
  - Arduino IDE (coding in C)
  - TinkerCAD (web-based, Circuits/3D Design)
  - Instructables (Report writing)
  - Arduino Forum
- All material is freely available
  - https://tiny.cc/mechatronix